Real-time Detection & Response System for Oidium neolycopersici and Similar Diseases using Electrochemical and VOC-based Sensing and Machine Learning Classification

Taha Rawjani & Matthew Li, Academies of Loudoun, Leesburg, Virginia, USA

BACKGROUND

Plant diseases cause approximately 20-40% of global crop losses annually, valued at around $220 billion. Current diagnostic methods like PCR and ELISA are unsustainable for field monitoring/diagnosis due to large costs and no way to monitor crops in real-time. By the time farmers see and diagnose their crops, it's too late: the damage would have already been done with the disease ravaging throughout the field.

Figure 1. United Nations goal of ending world hunger. A large contributor of food scarcity is plant disease (United Nations, 2015).

Specifically in the case of Powdery Mildew (a common plant pathogen), this would mean that your harvest of tomatoes, wheat, grapes or even apples would be toast as soon as its characteristic white powder appears.

Figure 2. Zinnias infected by powdery mildew (Photo taken by researchers).

Current ways for farmers to target Powdery Mildew is through the application of blanket pesticides, which are both bad for the environment (eutrophication/runoff) and costly ($3000 per hectare per month). Meanwhile, targeted control of these plants is significantly cheaper after the initial investment in remote sensing.
If these plant diseases can be detected and quarantined earlier, farming yield could significantly increase. This attacks the UN's sustainable development goal to end world hunger.

Solution Overview

01 Sensor Trained on VOC Data

02 Sensor Records Live Data

03 Live API & Real-time Updates

APPLICATIONS

- Deployment in agricultural and horticultural contexts where continuous, low-cost monitoring offers advantages over discrete laboratory assays, particularly commercial greenhouses producing high-value crops (tomato, cucumber, ornamental flowers).
- Distributed sensor networks provide early warning of powdery mildew, downy mildew, and Botrytis outbreaks, enabling localized fungicide application and infected-plant quarantine in place of prophylactic blanket spraying.
- Reduces pesticide expenditure (~$3,000/hectare/year) and mitigates environmental externalities including runoff-driven eutrophication and non-target organism toxicity.

ENGINEERING OBJECTIVE

Detect and classify plant diseases in real-time to prevent catastrophic famines.

Accurate	Replicable	Real-Time
Low error in classifying plants	*Easily scalable for farm coverage*	*Low latency in detection and update*

SOLUTION CRITERIA

(Icons from Flaticon)

SENSOR & DATA COLLECTION

Figure 3. A BME680 Sensor Connected to a ESP32 M5-Stick platform via a Grove cable, showcasing real-time environmental data measurements for Temperature, Humidity, Air Pressure, and Gas Resistance (Created by researchers).

Figure 4. A BME680 Sensor Connected to a ESP32 M5-Stack platform via a Grove cable, showcasing real-time environmental data measurements for Temperature, Humidity, Air Pressure, and Gas Resistance (Created by researchers).

Hardware Assembly

The BME680 sensor from Adafruit measures gas resistance, temperature, humidity, and air pressure. The most important metric being measured is Gas Resistance. When calibrated, this measurement can be used to determine the ppm (parts per million) concentration of VOCs. Unlike traditional MOS (Metal-Oxide Sensor) sensors, this one comes pre-calibrated with ready-to-use drivers. The sensor connects to an M5 Stick (ESP32 platform) via a Grove connector through soldering. The M5 Stick provides a display, wifi, and RTC, which is useful for timestamping and sending data to a server.

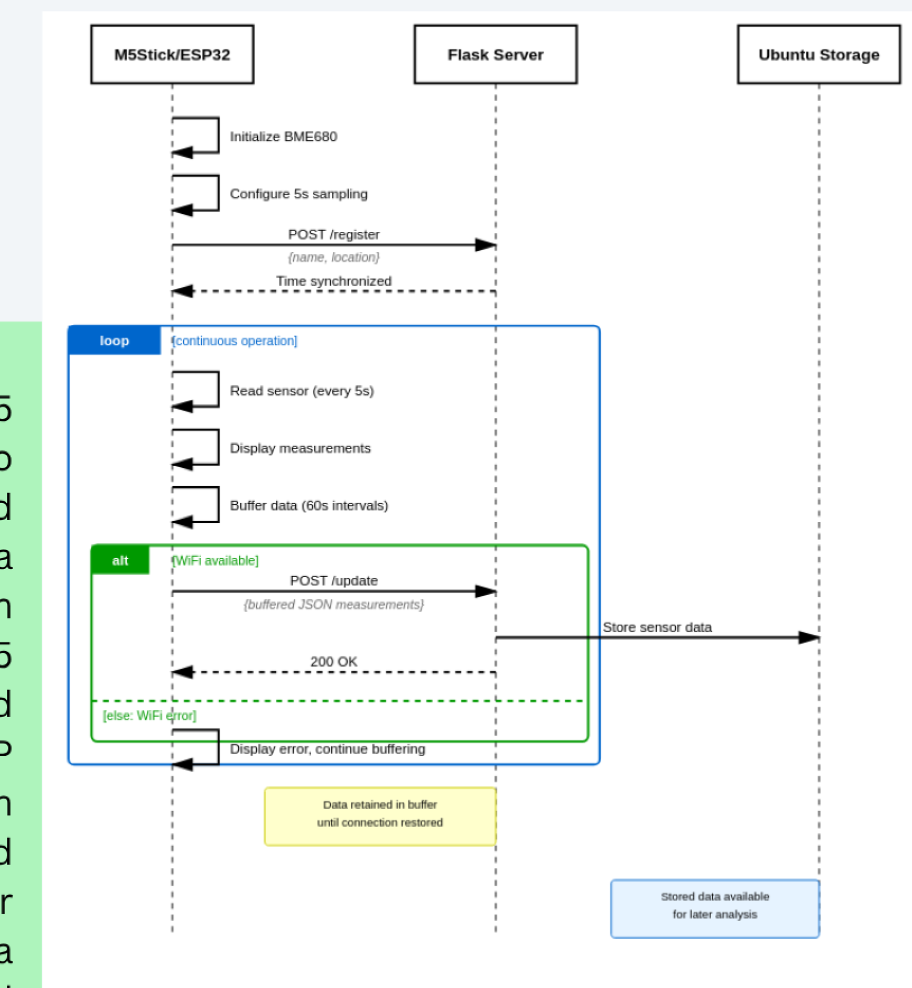

Figure 5. Sequence Diagram of the Proposed Sensor-Server Protocol with Server-Side Storage of Recorded Data.

Real-Time Server

The HTTP Server was written with Python, and it exposes an /update endpoint where the sensor sends buffered JSON updates with measurements. Upon initialization, the sensor uses the '/register' endpoint to identify itself with a name and location, and also synchronizes time with the server, allowing for the data sent by the sensor to be tracked and accurately identified. In the case that the connection to the server is not possible (e.g., due to a wi-fi error), the sensor was set to work normally with a small error display. In that case, the sensor was to buffer the data until a reliable internet connection could be established.

Software Integration

Embedded software (C++/Arduino/M5 Stick) frameworks were used to program the BME680. A 5-second sampling interval was configured with a graphical display for measurements on the M5 stick. The ESP32 (part of the M5 stick) buffered data into 60-second intervals and then transferred via HTTP POST to a Flask Server running on Ubuntu. This data collection method was established as a way to allow for long-term storage of sensor data on a server to be later used for analysis and processing.

DISEASE DIAGNOSIS & ANALYSIS

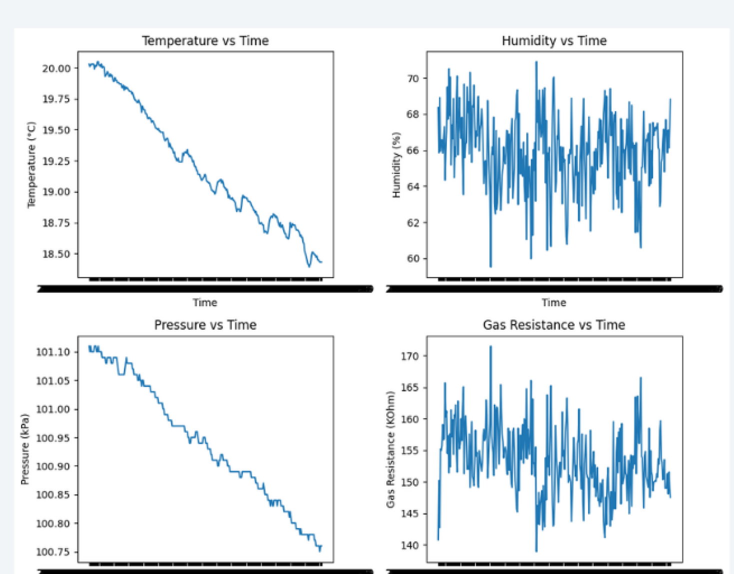

Figure 6. A graph of Temperature, Pressure, Humidity, and Gas Resistance vs. Time. Collected on a control group plant of Nephrophilis Fern for 3 consecutive days in an isolated environment.

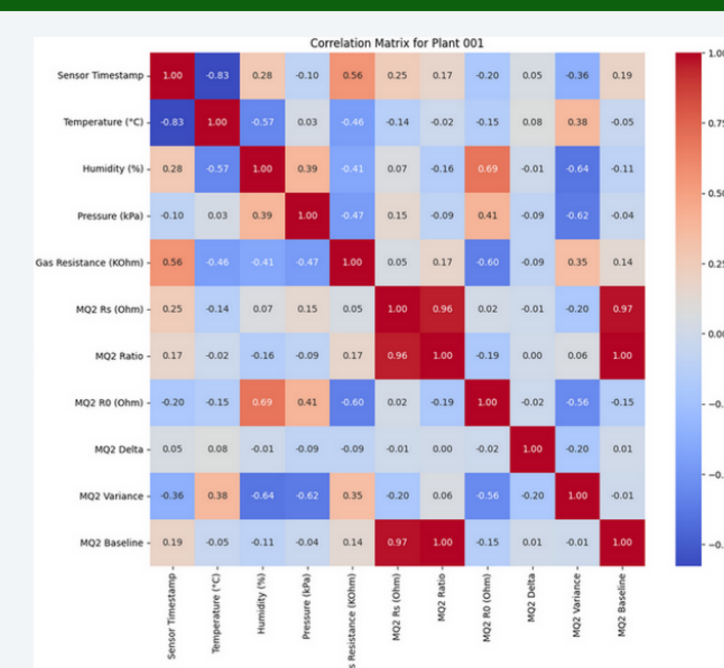

Figure 7. Correlation Matrix of features collected from the controlled plant.

- VOC emission from infected plants is well-established, with powdery mildew eliciting methyl salicylate, sesquiterpenes, and green leaf volatiles prior to visible symptoms — leveraged here via low-cost MOS sensing (BME680, MQ2) achieving 92% classification accuracy ($p < 0.01$).
- Methodological limitations constrain the diagnostic claim: single-element MOS sensors return a scalar total-VOC proxy susceptible to abiotic confounders (drought, mechanical damage, thermal fluctuation); $n = 3$ with within-plant temporal splits introduces pseudoreplication and likely inflates accuracy; reported gas resistance directionality warrants verification given the inverse relationship to reducing-gas concentration.
- Best characterized as a continuous low-cost anomaly detector for triggering downstream confirmatory assays (PCR, ELISA) rather than a standalone diagnostic; multi-element arrays, expanded replication, abiotic stress controls, and pre-symptomatic timeline characterization would strengthen the claim.

	Temperature (°C)	Humidity (%)	Pressure (kPa)	Gas Resistance (KOhm)	MQ2 Rs (Ohm)	MQ2 Ratio	MQ2 R0 (Ohm)	MQ2 Delta	MQ2 Variance	MQ2 Baseline
count	48	48	48	48	48	48	48	48	48	48
mean	20.661458	34.522917	101.416604	332.003125	743.572083	6.178498	274.4525	0.001837	0.385828	5.090817
std	1.942342	4.247135	0.522422	30.032939	2907.928223	8.942372	561.675722	0.060027	0.2816	8.944598
min	18.7	29.14	100.651	234.04	291.2	0.1684	48.68	-0.185	0	0.1293
25%	19.3	31.32	100.94025	311.2725	310.86	1.0525	48.68	-0.0149	0.023052	0.797025
50%	19.74	34.075	101.3595	345.23	320.08	6.4062	48.68	0.0008	0.439585	4.9999
75%	21.575	36.0975	101.847	355.0325	333.9875	6.68305	318.9	0.0324	0.546627	5.166125
max	25.04	52.95	102.296	360.28	20470	64.1895	2082.4	0.1409	1.171719	64.1895

Table 1. Centrality and Variability statistics for the control group plant.

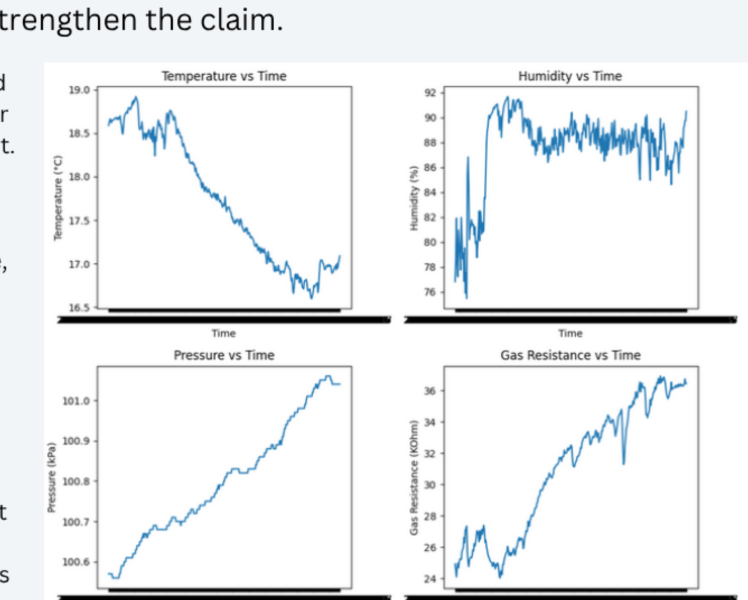

Figure 8. A graph of Temperature, Pressure, Humidity, and Gas Resistance vs. Time. Collected on an experimental group plant of Nephrophilis Fern for 3 consecutive days in an isolated environment. The plant was dehydrated and showed wrinkled leaves after three days.

DISCUSSION

i. Sensor Development and Engineering

The sensor was successfully developed and engineered after multiple iterations that built on each other. Firstly, a prototype using an Arduino Uno was built, but this was unable to measure VOCs as it didn't have a Grove port for the BME680 sensor. The Grove port is needed to convert the output from the BME680 into a digital signal that a microcontroller can understand, so a platform switch was performed from an Arduino platform onto the M5 Stick, using an ESP32.

However, a hardware limitation of the M5 Stick is that whenever it has Wi-Fi enabled, two of its three analog ports become disabled. This meant that it was impossible to connect my ESP32 to both the VOC sensors and the electrochemical sensor. Thus, for the data collection, electrochemical sensing was scrapped, and the project solely focused on VOC sensing in plants.

ii. Data Collection with Plants

Data was successfully collected from two groups of fern *necrophilis* plants: an experimental group with a disease, and a control group. It was analyzed using a variety of statistical tests, such as a T-test, yielding a p-value of 0.00, resulting in a statistically significant difference.

iii. Data Analysis and Modeling

The collected data was analyzed using machine learning. A random forest regression model was used to create a simple predictive model to determine the diseased status of a plant based on their VOC and electrochemical signals. A correlation matrix was constructed to find out similarities and correlations between collected features, as shown in Table 1 and Figure 3.

CONCLUSION

ADDRESSING CRITERIA

(Icons from Flaticon)

- Classification was seen as possible, but still requires more data to be fully replicable at a large scale.
- Extremely replicable, each sensor can take <5 hours to print and assemble, and costs less than $20.
- Cropsense's performance is real-time, with minimal processing delay measured and low server latency.

KEY REFERENCES

- S. Savary, L. Willocquet, S. J. Pethybridge, P. Esker, N. McRoberts, and A. Nelson, "The global burden of pathogens and pests on major food crops," Nature Ecology & Evolution, vol. 3, no. 3, pp. 430–439, 2019.
- D. A. Glawe, "The powdery mildews: a review of the world's most familiar (yet poorly known) plant pathogens," Annual Review of Phytopathology, vol. 46, pp. 27–51, 2008.
- T. J. Maginga, E. Masabo, P. Bakunzibake, K. S. Kim, and J. Nsenga, "Using wavelet transform and hybrid CNN–LSTM models on VOC & ultrasound IoT sensor data for non-visual maize disease detection," Heliyon, vol. 10, no. 4, p. e26781, 2024.
- N. Benkhaira, S. Haddad, M. Amamra, and A. Manseur, "Mint treatment day prediction using a multi-sensor system and machine learning algorithms," Sensors and Actuators A: Physical, vol. 328, p. 112782, 2021.
- Bosch Sensortec, "BME680: Low power gas, pressure, temperature & humidity sensor — data sheet," BST-BME680-DS001, 2017. [Online]. Available: https://www.bosch-sensortec.com/products/environmental-sensors/gas-sensors/bme680/
- J. K. Holopainen and J. Gershenzon, "Multiple stress factors and the emission of plant VOCs," Trends in Plant Science, vol. 15, no. 3, pp. 176–184, 2010.
- C. Rota, J. J. Carrami˜nana, J. Burillo, and A. Herrera, "In vitro antimicrobial activity of essential oils from aromatic plants against selected foodborne pathogens," Journal of Food Protection, vol. 71, no. 7, pp. 1499–1506, 2008.
- United Nations, "Transforming our world: the 2030 agenda for sustainable development," Resolution A/RES/70/1, Sep. 2015. [Online]. Available: https://sdgs.un.org/2030agenda